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COUNTRY: ISRAEL



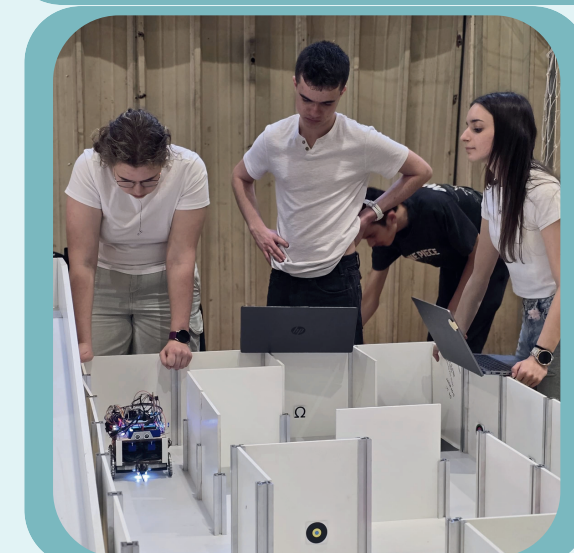
Jasmine Lvov
 Head of design, hardware, and electronics
 National OnStage Champion(2023-2025)
 European Vice OnStage Champion(2023)
 European OnStage Champion(2025)
 National Rescue Maze Champion (2026)



Matan Karaim
 Head of software
 National OnStage Champion(2023-2025)
 European Vice OnStage Champion(2023)
 European OnStage Champion(2025)
 National Rescue Maze Champion (2026)

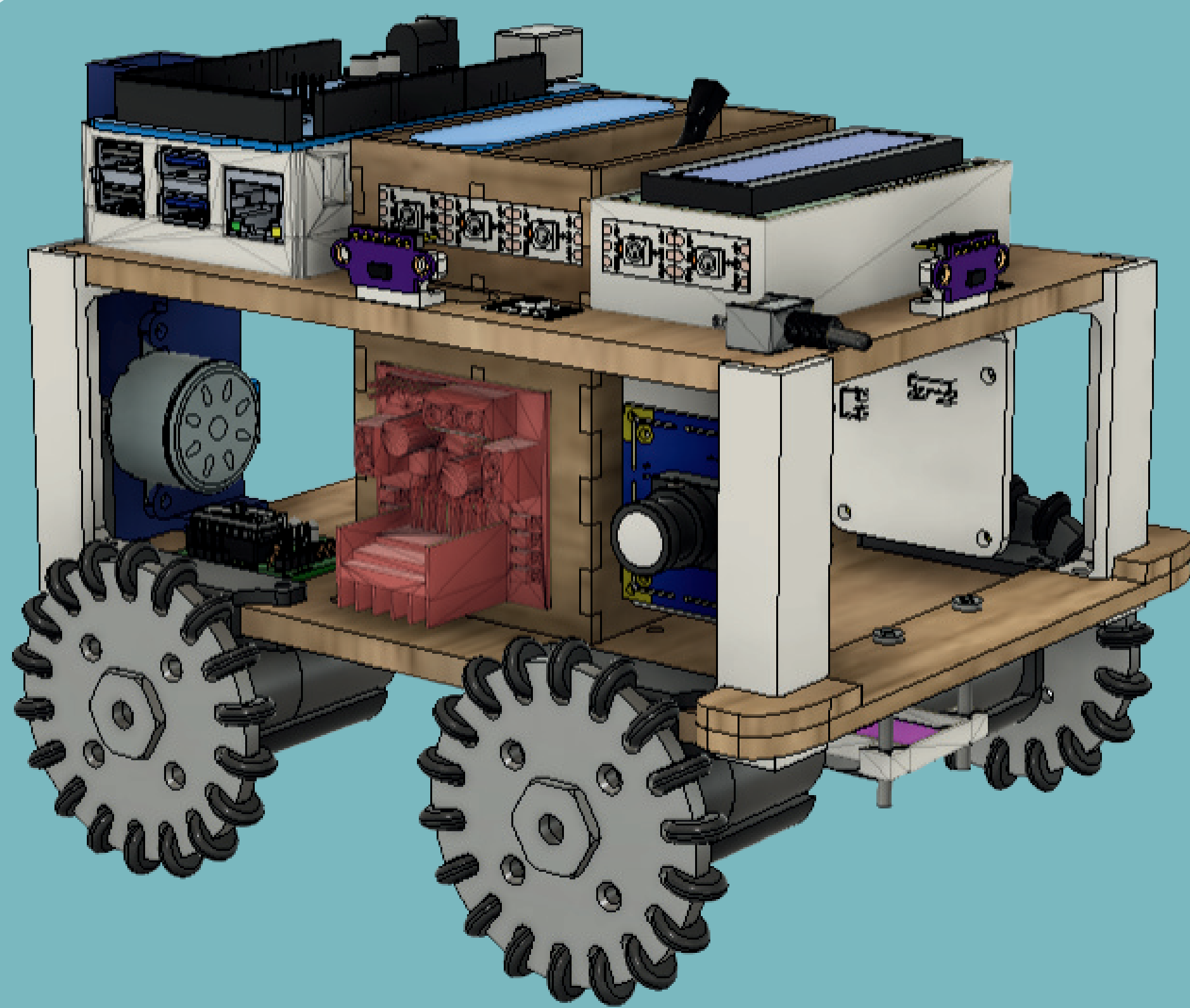
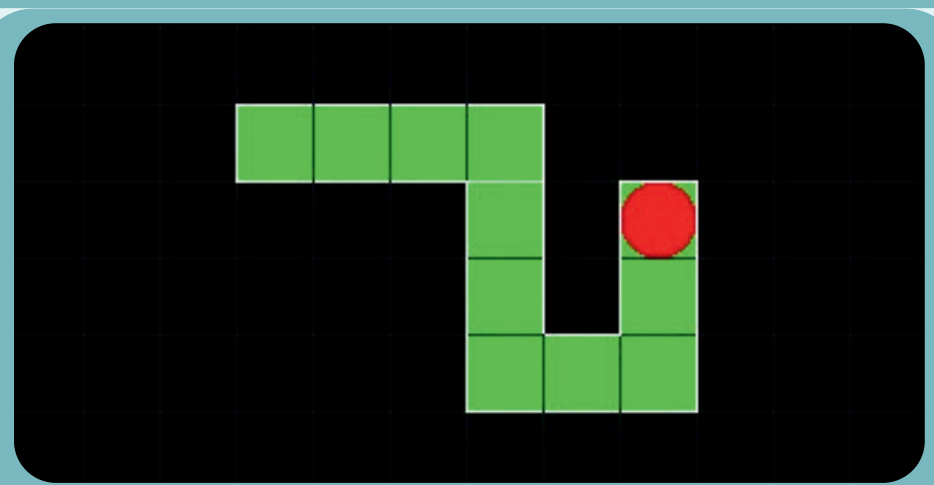


Noam Kalmar
 Head of victim recognition software
 National Soccer lightweight Entry Vice Champion (2022)
 National Soccer lightweight Entry Champion (2023)
 National OnStage Champion (2024-2025)
 European OnStage Champion(2025)
 National Rescue Maze Champion (2026)



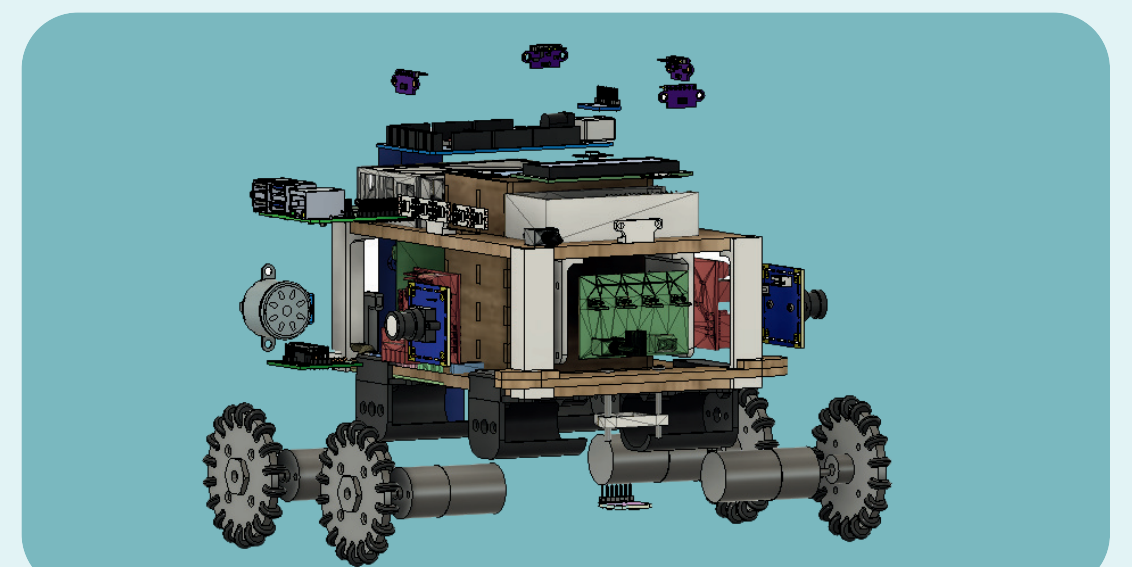
Victim recognition

All of the code related to the cameras, letter, and cognitive victim recognition doesn't run on the Arduino but on the Raspberry Pi. The raspberry searches both cameras for a potential letter or target. Once he has a potential, to be sure, he begins a scan in which he checks several times what the victims type. To recognize a letter victim, it searches for pixel sequences and compares them to a previously seen example of each of the three victim letters to check for a match. To find a cognitive target, it finds a circle and samples pixels from each layer to check for its color. In the end, when the Raspberry is sure of the victim's type, it sends the Arduino the appropriate serial message so he could react accordingly.



Arduino and Raspberry Pi Communication

The robot's main processor is the Arduino Mega 2560, which uses C++ and is in charge of receiving and analyzing information from the sensors, controlling the motors, navigating the maze, and dropping the rescue kits. The secondary processor is the Raspberry Pi, that runs Python; it's in charge of processing the images received from the cameras. When it recognizes a victim, it sends the Arduino a message via serial communication, and the robot reacts accordingly.



Navigation and Mapping

The robot navigates the maze by using real-time sensor feedback with dynamic mapping and a Breadth First Search (BFS) algorithm. The robot has four Time-of-Flight (ToF) distance sensors and a BNO055 gyroscope, which allows it to know how much distance he passed, straighten in the corridors, and make turns. At each new tile, the distance sensors scan for walls, constantly updating the robot's internal map. If the robot detects an open, unvisited adjacent tile, it chooses the direct path and moves towards it. However, when the robot encounters a dead end or finds itself surrounded only by walls and already visited tiles, it initiates the BFS algorithm to determine its next move based on the map generated so far. The algorithm calculates the shortest way from the robot's current position to all accessible cells, effectively identifying the closest unvisited tile in the entire maze. Once this target is selected, the algorithm computes the fastest route to get there, allowing the robot to efficiently backtrack and continue its exploration without getting stuck.

Hardware

The robot is fully designed in Fusion360 and brought to life using laser cutting and 3d printing.

The robot uses a combination of an **Arduino** and a **Raspberry Pi 4** since the Arduino isn't strong enough to process the image received from the **camera**.

For navigating in the maze, the robot uses a combination of **VL53L0X ToF**, **TCS34725** color sensors, and **Bno055 imu** to detect the walls, the different colored tiles, and stay aligned in the middle of the corridor.

For driving, the robot uses 4 **25ga-370** (with the **L298N** motor driver) and 4 **Omni wheels**. In the past, we tried making our own wheels, but since they were 3d printed and we didn't manage to make them have enough grip, we switched to Omni wheels.

To indicate the current state of the robot and if victims are found, we have an **LCD screen** and 2 **LED strips**, one for the victim type and one for the current status, with 4 LEDs to mark where the walls are.

The robot is powered by a **portable charger** for the Raspberry Pi and a **12V lithium battery** for the rest of the robot. They are placed between the levels so that they don't throw off the center of mass.

The kit deployment mechanism uses a **stepper motor** with a kick leg, and the kits sit in an L-shaped tube to avoid accidentally losing kits while driving or having too many kits fall when treating a victim.

To avoid accidentally getting stuck in the gaps between the walls the robot has a rounded front extension.

